Clustered Multiple Manifold Metric Learning for Hyperspectral Image Dimensionality Reduction and Classification

Yanni Dong, *Member, IEEE*, Yao Jin, and Shunbo Cheng

***Abstract*— Dimensionality reduction (DR) technology is an important part of hyperspectral image (HSI) processing. The DR technology can effectively remove the redundant information in the HSIs and avoid the Hughes phenomenon, which is beneficial to image classification. Metric learning is widely used in DR technology, the goal of which is to achieve DR by maximizing the distance of between-class while minimizing the distance of within-class. However, traditional metric learning does not consider the intrinsic structure of data during the training process, resulting in insufficient information utilization. In order to solve the above problems, this article proposes clustered multiple manifold metric learning (CM^{3}L) by combining manifold learning with metric learning for DR and classification of HSI. The proposed CM^{3}L algorithm first divides the original data into multiple independent clusters and regards each sample point and its near point in the cluster as a whole, constructing it as a manifold. Then, CM^{3}L uses the manifold metric distance to replace the traditional metric distance. Finally, by making full use of the local information of HSI in such a way, the discrimination ability is enhanced. Intensive experimental results on three real HSI datasets validate the effectiveness of our proposed CM^{3}L algorithm.**

***Index Terms*— Dimensionality reduction (DR), hyperspectral image (HSI), manifold learning, metric learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain hundreds of spectral bands, which provide both the detailed spatial structure and spectral information. Therefore, HSI can distinguish the subtle differences of ground cover classes and are widely used in the fields of atmospheric monitoring, vegetation monitoring, disaster warning, and so on [1]–[7]. However, the large amount of spectral bands also contains great redundancy that not only increases the computational resources but also causes the emergence of the Hughes phenomenon (that is, when the number of features of the input data exceeds a certain critical point, further increasing feature data will lead to the reduction of classification accuracy) [8]–[13]. In order to address this issue, dimensionality reduction (DR) methods, which can preserve valid information while reducing the dimension of feature space, have been applied to HSI processing [14]–[18].

According to the situation of the existence of prior sample information, DR methods can be divided into two categories: supervised and unsupervised approaches. Supervised algorithms use the information provided by prior samples to complete DR, while unsupervised algorithms do not apply any prior information. The typical supervised algorithms include the linear discriminant analysis (LDA) [19], [20], which aims to find a projection matrix that maximizes the between-class scatter and minimizes the within-class scatter. The principal component analysis (PCA) [21], [22] finding the subspace of orthogonal projection by maximizing the variance of the input matrix is the most widely used unsupervised algorithm. Both LDA and PCA are developed on the basis of the statistical theory, which assumes that data satisfy the statistical distribution and cannot consider the intrinsic structure of the data [23], [24]. Therefore, in the face of complex data, the statistics-based algorithm cannot achieve satisfactory results.

In essence, DR learning looks for low-dimensional representation of high-dimensional feature space by assuming the Gaussian model and discriminates within-class and between-class samples in the new feature space. Metric learning can utilize prior sample information to minimize the distance of the same class and maximize the distance of different classes to achieve the separation of different categories, which meets the requirements of DR [25], [26]. Thus, many metric learning-based methods are used in the field of DR. For example, large margin nearest neighbor (LMNN) [27] makes the nearest neighbors around the sample point of the same class with the sample point through training, while the sample point of different classes is separated far away. The neighborhood component analysis (NCA) [28] aims to select instances of the same kind as its neighbors with the largest sum of probabilities; information-theoretic metric learning (ITML) [29] converts the Mahalanobis metric learning into

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solving the Bregman optimization problem by minimizing the differential relative entropy between two multivariate Gaussian distributions; and locally adaptive dimensionality reduction metric learning (LADRml) [30] uses local adaptive constraints to relax the fixed threshold and adaptively makes classification decisions based on changes in the threshold and the distance learned by the metric. These algorithms based on global metric learning will only calculate a metric matrix in the end. When faced with heterogeneously distributed data, it is not enough to obtain satisfactory results. To solve the above problem, some multiple local metric learning algorithms are proposed [31], [32], for example, clustered multimetric learning (CMML) [33], LMNN with multiple metrics (mmLMNN) [27], and so on. Similarly, both global metric learning methods and multiple local metric learning methods ignore the structural characteristics of the data and cannot exploit the intrinsic structure information. This means that the algorithm does not make full use of the information provided by the training samples. However, the development of manifold learning solves the above problems well.

The manifold learning algorithm, which aims to find out the low-dimensional manifold structure embedded in the high-dimensional space in the data and find out the inherent law of data distribution, is a nonlinear algorithm developed in recent years [34]–[39]. Representative algorithms for manifold learning are Laplacian eigenmap (LE) [40] and local linear embedding (LLE) [41]. LE is based on spectral graph theory, which aims to use a graph embedding algorithm to preserve the nearest neighbor structure of data. LLE calculates the low-dimensional representation of data by preserving the linear structure of the original high-dimensional data neighborhood. However, those algorithms are confronted with the out-of-sample problem [42], so the linear approximation technique is introduced into the nonlinear manifold learning algorithms. Neighborhood preserving embedding (NPE) [43], [44] is a linear version of LLE and locality preserving projection (LPP) [45] linearly approximate to LE. In addition, some studies combine manifold learning with other methods, which have achieved good results. For example, by integrating LPP and LDA, the local Fisher discriminant analysis (LFDA) [46] is proposed, which preserved the structure of multimodal non-Gaussian class distribution. Local constrained manifold structure collaborative preserving embedding (LMSCPE) [47] combines manifold learning with graph learning, which can well characterize the collaborative relationship between sample pairs and explore the intrinsic geometric structure in HSI. Local manifold-based sparse discriminant learning (LMSDL) [38] presents a manifold-based sparse relationship between samples, which can reveal the manifold structure and complex sparse relationships of HSI data, and significantly enhance the HSI classification capability.

In addition, in the field of face recognition, some scholars have proved that a face image is a manifold, and then, the distance between different manifolds can be calculated [48]–[50]. By maximizing the distance between heterogeneous manifolds and minimizing the same kind of manifolds, the distance can be distinguished as different classes, such as the log-Euclidean metric learning (LEML) algorithm [18] and the projection metric learning (PML) algorithm [51]. Both algorithms use the symmetric positive definite (SPD) matrix to represent the manifold and then use the geodesic distance to calculate the manifold distance. The difference between the two algorithms is that, when calculating the manifold distance, LEML performs tangential space transformation on the original data, while PML uses the orthogonal basis matrix of the original data for calculation. However, the above two algorithms both use the geodesic distance when calculating the manifold distance. The feature space obtained in the final training is also based on the geodesic distance, and the Euclidean distance is often used as the basis in the classification process, which may be a negative impact on classification results [52].

Motivated by the above discussion, in order to overcome the aforementioned problem and enhance the discriminant ability by exploiting intrinsic structure information, this study proposes the clustered multiple manifold metric learning (CM^3L) for DR and classification of HSI. CM^3L combines with manifold learning on the basis of CMML, constructing in a manifold for each sample point in the training sample and its nearest neighbors, and using the manifold distance for training. The final goal is to minimize the within-class manifold distance and maximize the between-class manifold distance for achieving a better performance of classification.

The main contributions of this article are given as follows.

1) In order to deal with complex nonlinear data, the proposed algorithm uses a divide-and-conquer strategy to divide the original data into multiple clusters, in each of which a distance metric is trained to separate the data locally for effectively training the details of data.
2) To make full use of limited prior information and enhance the discrimination ability, the proposed algorithm combines the idea of manifold learning with metric learning, constructing a manifold by taking the sample point as the center and combining the neighboring points.
3) Considering the different contributions of each neighboring point around the sample point, the proposed algorithm uses different weights to exploit local information by evaluating each neighboring point for ensuring that neighboring points closer to the sample point have greater weight.

This article is organized as follows. Section II briefly reviews some related works, including metric learning and CMML. Section III details the proposed CM^3L method. The experimental results on three real hyperspectral datasets are presented in Section IV to demonstrate the effectiveness of CM^3L. Finally, Section V provides some concluding remarks and suggestions for future work.

II. Related Works

A. Metric Learning

For HSI, we use matrix $X = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n] \in R^{d\times n}$ to represent the training dataset, where d represents the number of bands and n represents the number of pixels. $Y = [y_1, y_2, \ldots, y_n] \in R^{1\times n}$ is the associated labels.

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According to the label information, the training set can be divided into similar constraints set S and dissimilar constraints set D as

$$S : \forall(\mathbf{x}_i, \mathbf{x}_j)\,|_{y_i=y_j} \in S \tag{1}$$

$$D : \forall(\mathbf{x}_i, \mathbf{x}_l)\,|_{y_i\neq y_l} \in D. \tag{2}$$

The goal of metric learning is to get a Mahalanobis distance metric matrix $\mathbf{M}$, which is the symmetric and positive semidefinite matrix and satisfies the triangle inequality properties [53], [54]. The distance $d_{\mathbf{M}}(\mathbf{x}_i, \mathbf{x}_j)$ between $\mathbf{x}_i$ and $\mathbf{x}_j$ can be computed as

$$d_{\mathbf{M}}(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{(\mathbf{x}_i - \mathbf{x}_j)^T \mathbf{M}(\mathbf{x}_i - \mathbf{x}_j)} \tag{3}$$

where $(.)^T$ represents the transpose of the matrix. Based on the Mahalanobis distance, the target matrix $\mathbf{M}$ can be obtained by maximizing the between-class distance and minimizing the within-class distance

$$\arg\min_{\mathbf{M}} \left(\sum_{\mathbf{x}_i, \mathbf{x}_j \in S} d_{\mathbf{M}}(\mathbf{x}_i, \mathbf{x}_j) - \sum_{\mathbf{x}_i, \mathbf{x}_l \in D} d_{\mathbf{M}}(\mathbf{x}_i, \mathbf{x}_l) \right). \tag{4}$$

In order to deal with DR, we can apply the transformation of $\mathbf{M}$ to get a nonsquare matrix $\mathbf{W} \in R^{d\times m} (m \ll d)$, which maps data from the original high-dimensional space into the low-dimensional space such that

$$\mathbf{M} = \mathbf{W}\mathbf{W}^T. \tag{5}$$

To jointly undertake DR and metric learning, (3) and (5) can be combined as

$$\begin{aligned} d_{\mathbf{M}}(\mathbf{x}_i, \mathbf{x}_j) &= \sqrt{(\mathbf{x}_i - \mathbf{x}_j)^T \mathbf{W}\mathbf{W}^T (\mathbf{x}_i - \mathbf{x}_j)} \\ &= \sqrt{\left(\mathbf{W}^T\mathbf{x}_i - \mathbf{W}^T\mathbf{x}_j\right)^T \left(\mathbf{W}^T\mathbf{x}_i - \mathbf{W}^T\mathbf{x}_j\right)} \\ &= \left\|\mathbf{W}^T(\mathbf{x}_i - \mathbf{x}_j)\right\|. \end{aligned} \tag{6}$$

Different metric learning algorithms can get the metric matrix $\mathbf{M}$ (or $\mathbf{W}$) by constructing different constraints under the supervision of set S and D, which can reduce the dimensions and distinguish different samples in a low-dimensional space [55]–[57].

B. CMML

All the abovementioned are global metric learning, and when faced with relatively complex nonlinear data, multiple local metric learning, such as CMML, can achieve satisfactory results, which divides the training samples into multiple clusters, and the training task of each cluster can be regarded as an independent global metric learning.

For CMML, the k-means algorithm is used to divide the training samples into U mutually independent clusters $X = C^{(1)} \cup \cdots \cup C^{(U)}$. That is, each training sample only belongs to one cluster. For each cluster, we have to learn a metric matrix that satisfies the triple constraint as

$$\begin{aligned} \tau^{(c)} = \big\{ (\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_l) | d_{\mathbf{M}^{(c)}}(\mathbf{x}_i, \mathbf{x}_j) < d_{\mathbf{M}^{(c)}}(\mathbf{x}_i, \mathbf{x}_l) \\ \mathbf{x}_i, \mathbf{x}_j \in S, \mathbf{x}_i, \mathbf{x}_l \in D \big\}. \end{aligned} \tag{7}$$

For any triple constraint $(\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_l \in \tau^{(c)})$, $1 \le c \le U$, it can be achieved by maximizing the margin as

$$d^2_{\mathbf{M}^{(c)}}(\mathbf{x}_i, \mathbf{x}_l) - d^2_{\mathbf{M}^{(c)}}(\mathbf{x}_i, \mathbf{x}_j) = \left\langle \mathbf{M}^{(c)}, Z_r \right\rangle \tag{8}$$

where $Z_r = (\mathbf{x}_i - \mathbf{x}_l)(\mathbf{x}_i - \mathbf{x}_l)^T - (\mathbf{x}_i - \mathbf{x}_j)(\mathbf{x}_i - \mathbf{x}_j)^T$. Many loss functions can be used to maximize the margin, such as the square or logistic loss. CMML minimizes the hinge loss as

$$\max\left(1 - \left\langle \mathbf{M}^{(c)}, Z_r \right\rangle, 0\right). \tag{9}$$

In addition, CMML avoids overfitting by setting up a global metric matrix to propagate information between clusters. Let $\mathbf{M}^{(0)} \succ 0$ denote the global metric matrix. Then, the squared Frobenius norm is used to make each local metric matrix $\mathbf{M}^{(c)}$ similar to the global metric matrix $\mathbf{M}^{(0)}$ as

$$\left\|\mathbf{M}^{(0)} - \mathbf{M}^{(c)}\right\|_F^2, \quad 1 \le c \le U. \tag{10}$$

In this way, all the local metric matrices can be related to each other, improving the generalization ability [58]. Finally, CMML adds a trace-norm regularization to the global metric matrix $\mathbf{M}^{(0)}$, which imposes a low-rank constraint to $\mathbf{M}^{(0)}$. Similarly, because the local metric matrix $\mathbf{M}^{(c)}$ is similar to the global metric matrix $\mathbf{M}^{(0)}$, the low-rank constraint will also be imposed on the local metric matrix $\mathbf{M}^{(c)}$. In this way, overfitting can be reduced. According to the abovementioned, we can get the objective function of CMML as

$$\begin{aligned} \arg\min_{\mathbf{M}^{(0)},\ldots,\mathbf{M}^{(U)}} \quad & \alpha \mathrm{tr}\left(\mathbf{M}^{(0)}\right) \\ & + \frac{1}{U}\sum_{c=1}^{U}\left[\frac{\beta}{2}\left\|\mathbf{M}^{(0)} - \mathbf{M}^{(c)}\right\|_F^2 + \frac{1}{N^{(c)}}\sum_{r\in\tau^{(c)}}\xi_r\right] \\ \text{s.t.} \quad & \forall r \in \tau^{(c)}, \quad c \in \{1, \ldots, U\} : \\ & \left\langle \mathbf{M}^{(c)}, Z_r \right\rangle \ge 1 - \xi_r, \quad \xi_r \ge 0 \\ & \mathbf{M}^{(0)}, \ldots, \mathbf{M}^{(U)} \succ 0 \end{aligned} \tag{11}$$

where $N^{(c)}$ is the number of constraints in $\tau^{(c)}$. $\alpha > 0$ and $\beta > 0$ are hyperparameters, which are used to balance the effects of various parts. ξ_r is the slack variables, which is used to deal with the influence of outliers on the results.

III. Proposed Method

Though CMML learns multiple distance measures to process heterogeneous distributed data, it cannot consider the intrinsic structure of the data itself. Therefore, we refer to the idea of manifold learning and take each sample point as the center, regarding the sample point and its surrounding neighboring points as a manifold to minimize the within-class manifold distance while maximizing the between-class manifold distance. Based on the above ideas, we propose the CM^{3}L algorithm. Fig. 1 shows the flowchart of the proposed algorithm. As shown in the figure, we first use the k-means algorithm to divide the training samples into multiple clusters, then treat each cluster as a separate training task, and train the samples in each cluster. For each training sample in the

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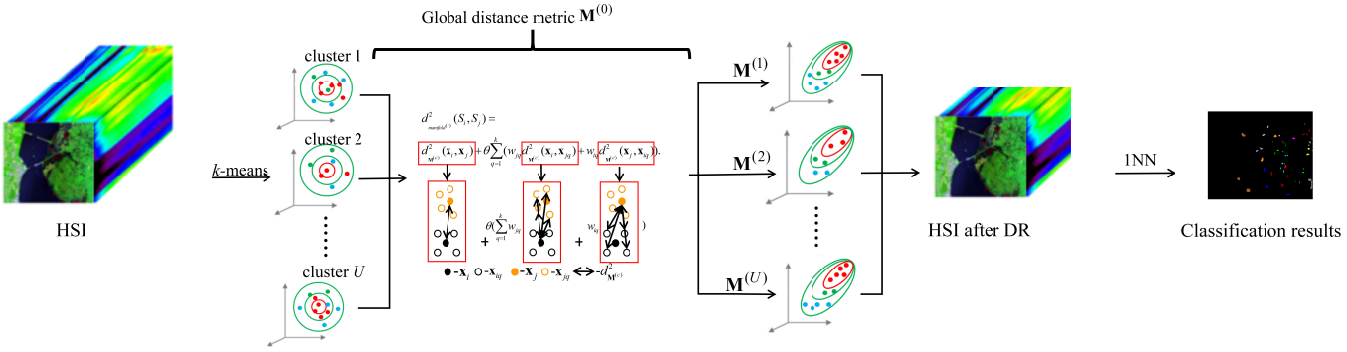


Fig. 1. Flowchart of the proposed algorithm.

cluster, we first find the nearest k neighboring points to construct a manifold and then calculate the distance between the manifolds instead of the Mahalanobis distance in traditional metric learning. In addition, a global metric matrix is set up to associate the local metric matrix obtained from each cluster to avoid overfitting.

A. Manifold Distances

The traditional metric learning method is generally based on the Mahalanobis distance, which can directly obtain the distance between two points and evaluate the similarity between two points. However, this model does not take into account the domain information around the sample points, resulting in insufficient use of information, and it is also vulnerable to the negative effects of noise and extreme points. In this article, the proposed manifold distance treats the sample and its surrounding neighboring points as a manifold. When obtaining the manifold to manifold distance, each point in the manifold will contribute to the calculation of the distance. In this way, while exploiting intrinsic structure information, it can also effectively reduce the impact of noise and extreme data. The specific calculation steps are given as follows. For each training sample $\mathbf{x}_i$ and its nearest k similar points, we regard it as a manifold $S_i = [\mathbf{x}_i, \mathbf{x}_{i_1}, \ldots, \mathbf{x}_{i_k}] \in R^{d\times(k+1)}$. k is the number of similar neighbors. The manifold distance metric can be formulated as

$$d^2_{\text{manifold}^{(c)}}(S_i, S_j) = d^2_{\mathbf{M}^{(c)}}(\mathbf{x}_i, \mathbf{x}_j) + \theta \sum_{q=1}^{k} \left(d^2_{\mathbf{M}^{(c)}}(\mathbf{x}_i, \mathbf{x}_{jq}) + d^2_{\mathbf{M}^{(c)}}(\mathbf{x}_j, \mathbf{x}_{iq})\right) \quad (12)$$

where the first item is to calculate the distance between the center points of the manifold, which is equivalent to the way to calculate the distance between sample points in traditional metric learning. The second item is to calculate the distance between other points in the manifold and the center points of another manifold on the basis of traditional metric learning. In addition, the parameter θ is to balance the effect of the two parts on the results.

B. Weighting Function

However, the adjacent points of the sample provide different information in the process of constructing the manifold. In order to avoid ignoring those adjacent points that provide more information and find complementary information, we use a weighting function to measure the similarity between the sample and its neighbors, which is more robust than the pairwise Euclidean distance. The weight matrix can be expressed as a linear combination of adjacent neighborhoods and combination coefficients w_{iq}, which can be used to measure the similarity of samples. The calculation formula is given as follows:

$$\arg\min_{w} \sum_{i=1}^{N} \left\| \mathbf{x}_i - \sum_{q=1}^{k} w_{iq}\mathbf{x}_{iq} \right\|. \quad (13)$$

The coefficient w_{iq} represents the contribution made by the neighboring points of the sample in the reconstruction of the sample. If the sample is more similar to the neighboring points, the contribution of the neighboring points during the reconstruction process will be greater, and the corresponding weight will be greater. The reverse is equally true. Thus, formula (12) can be rewritten as

$$d^2_{\text{manifold}^{(c)}}(S_i, S_j) = d^2_{\mathbf{M}^{(c)}}(\mathbf{x}_i, \mathbf{x}_j) + \theta \sum_{q=1}^{k} \left(w_{jq} d^2_{\mathbf{M}^{(c)}}(\mathbf{x}_i, \mathbf{x}_{jq}) + w_{iq} d^2_{\mathbf{M}^{(c)}}(\mathbf{x}_j, \mathbf{x}_{iq})\right). \quad (14)$$

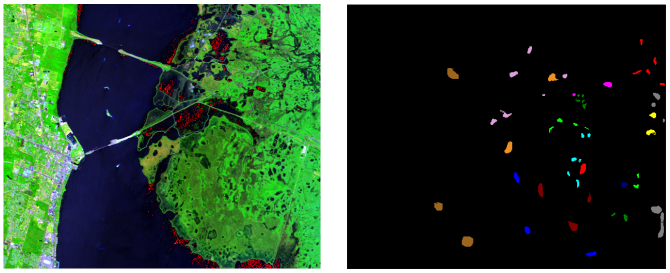
Using formula (14) to replace the distance formula in formula (8), we can get

$$d^2_{\text{manifold}^{(c)}}(S_i, S_l) - d^2_{\text{manifod}^{(c)}}(S_i, S_j) = \left\langle \mathbf{M}^{(c)}, Z_r^M \right\rangle. \quad (15)$$

Then, formula (11) can be rewritten as

$$\begin{aligned} \arg\min_{\mathbf{M}^{(0)},\ldots,\mathbf{M}^{(U)}} \quad & \alpha \text{tr}\left(\mathbf{M}^{(0)}\right) \\ & + \frac{1}{U}\sum_{c=1}^{U}\left[\frac{\beta}{2}\left\|\mathbf{M}^{(0)} - \mathbf{M}^{(c)}\right\|_F^2 + \frac{1}{N^{(c)}}\sum_{r\in\tau^{(c)}}\xi_r\right] \\ \text{s.t.} \quad & \forall r \in \tau^{(c)}, \quad c \in \{1, \ldots, U\}: \\ & \left\langle \mathbf{M}^{(c)}, Z_r^M \right\rangle \geq 1 - \xi_r, \quad \xi_r \geq 0 \\ & \mathbf{M}^{(0)}, \ldots, \mathbf{M}^{(U)} \succ 0. \end{aligned} \quad (16)$$

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(a) (b)

Fig. 2. (a) False-color image of the KSC dataset. (b) Ground-truth map.

C. Optimization Solver

In order to obtain the optimal result, we solve the following problems through stochastic subgradient descent (SGD):

$$\begin{aligned}&\arg\min_{\mathbf{M}^{(c)}} \frac{\beta}{2}\left\|\mathbf{M}^{(0)}-\mathbf{M}^{(c)}\right\|_F^2+\frac{1}{N^{(c)}}\sum_{r\in\tau^{(c)}}\xi_r\\&\text{s.t.}\quad \forall r\in\tau^{(c)}:\\&\qquad\left\langle\mathbf{M}^{(c)},Z_r^M\right\rangle\geq 1-\xi_r,\quad \xi_r\geq 0\\&\qquad\mathbf{M}^{(c)}\succ 0.\end{aligned}\tag{17}$$

First, we rewrite formula (17) into the following form so that the objective function is the sum of the loss functions associated with each triplet constraint:

$$\begin{aligned}&F\left(\mathbf{M}^{(c)}\right)\\&=\frac{\beta}{2}\left\|\mathbf{M}^{(0)}-\mathbf{M}^{(c)}\right\|_F^2+\frac{1}{N^{(c)}}\sum_{r\in\tau^{(c)}}\max\left(1-\left\langle\mathbf{M}^{(c)},Z_r^M\right\rangle,0\right).\end{aligned}\tag{18}$$

At the tth iteration, we replace the objective function $F(\mathbf{M}^{(c)})$ with an approximation based on a single triplet constraint $r\in\tau^{(c)}$ as

$$f_t\left(\mathbf{M}^{(c)}\right)=\frac{\beta}{2}\left\|\mathbf{M}^{(0)}-\mathbf{M}^{(c)}\right\|_F^2+\max\left(1-\left\langle\mathbf{M}^{(c)},Z_r^M\right\rangle,0\right).\tag{19}$$

Then, we calculate the subgradient at $\mathbf{M}_t^{(c)}$, given by

$$\nabla_t^{(c)}=\beta\left(\mathbf{M}_t^{(c)}-\mathbf{M}_t^{(0)}\right)-\left[\left[\left\langle\mathbf{M}^{(c)},Z_r^M\right\rangle<1\right]\right]Z_r^M\tag{20}$$

where [[.]] denotes that, when the condition is true, the value is 1; otherwise, it is 0. Then, we update $\mathbf{M}_t^{(c)}$ to $\mathbf{M}_{t+1}^{(c)}$ by setting

$$\begin{aligned}\mathbf{M}_{t+1/2}^{(c)}&=\left(1-\frac{1}{t}\right)\mathbf{M}_t^{(c)}+\frac{1}{t}\mathbf{M}^{(0)}+\eta_t\left[\left[\left\langle\mathbf{M}^{(c)},Z_r^M\right\rangle<1\right]\right]Z_r^M\\\mathbf{M}_{t+1}^{(c)}&=\mathbf{M}_{t+1/2}^{(c)}-\min(\lambda_{\min},0)u_{\min}u_{\min}^T\end{aligned}\tag{21}$$

where $\lambda_{\min}$ is the smallest eigenvalue of $\mathbf{M}_{t+1/2}^{(c)}$ with the corresponding eigenvector $u_{\min}$. Algorithm 1 illustrates the procedure of the proposed CM^{3}L algorithm.

IV. Experiments

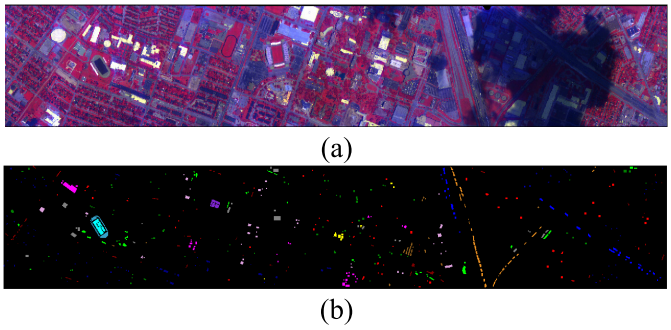
In this section, we conduct experiments on three real HSI datasets to validate the effectiveness of the proposed algorithm and select several state-of-the-art methods for comparison.

Algorithm 1 Procedure of the Proposed CM^{3}L Algorithm

Input: Training samples: $\mathrm{X}=[\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_n]\ \in \mathrm{R}^{d\times n}$
Training labels: $Y=[y_1,y_2,\ldots,y_n]\ \in R^{1\times n}$
Parameters: k, α, β, θ, $\mathbf{M}^{(0)}$

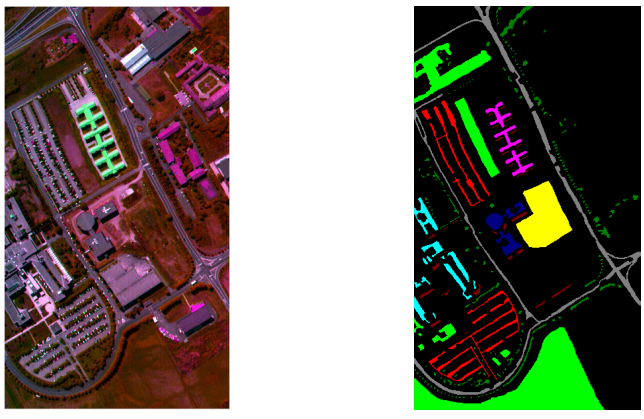
Steps: Use k-means to divide training samples into C clusters, and perform the following operations on the data in each cluster.
For each training sample $\mathbf{x}_i$, do
1). Select k nearest neighbors for $\mathbf{x}_i$
2). Construct the manifold S_i for $\mathbf{x}_i$
3). Calculate the coefficient ω_{iq} according to Formula (13).
4). Calculate $d^2_{manifold^{(c)}}(X_i,X_j)$ according to Formula (14).
5). Solve the optimization problem in Formula (16).

Output: Local metric matrix $\mathbf{M}^{(c)}$.



(a) (b)

Fig. 3. (a) False-color image of the Houston dataset. (b) Ground-truth map.



(a) (b)

Fig. 4. (a) False-color image of the Pavia University dataset. (b) Ground-truth map.

A. Dataset Description

The first dataset, the KSC dataset [59], was acquired by NASA's Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) instrument at the Kennedy Space Center (KSC) in Florida on March 23, 1996. This dataset contains 512 scan lines, and each line has 614 pixels. AVIRIS acquires data in 224 bands of 10-nm width with center wavelengths from 400 to 2500 nm. The KSC data, acquired from an

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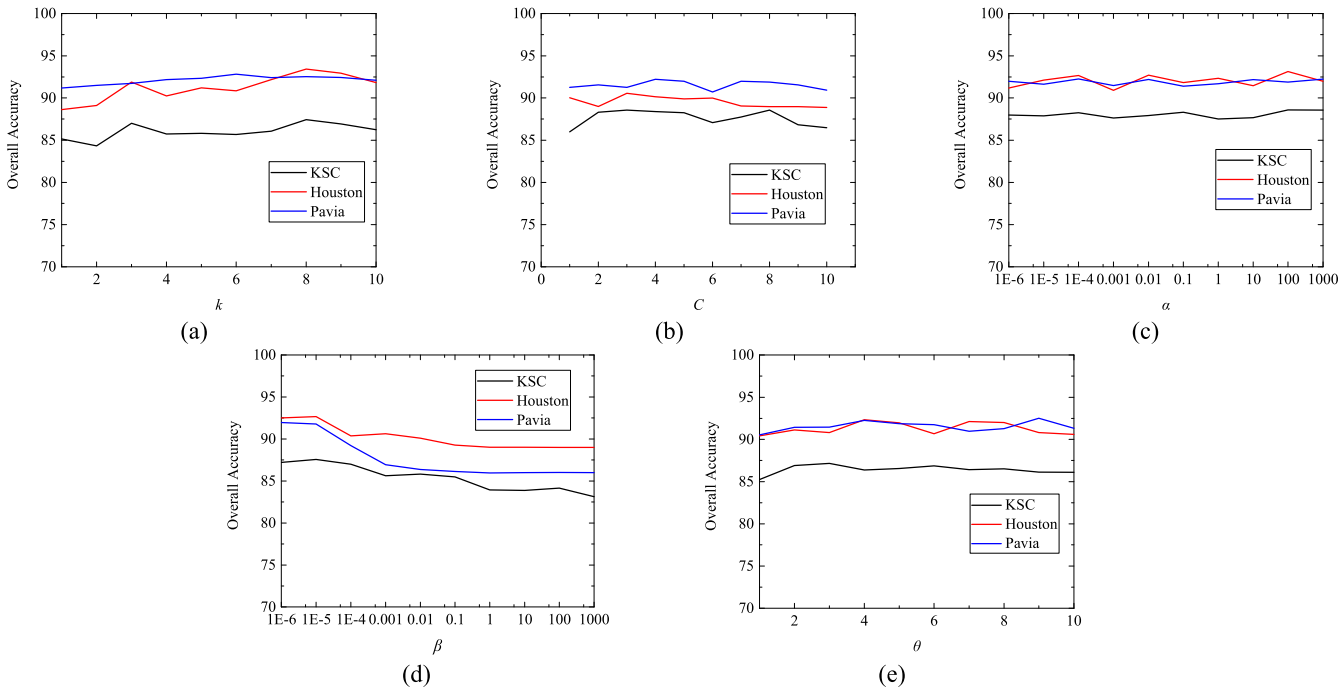


Fig. 5. Classification results of different parameters. (a) k. (b) C. (c) α. (d) β. (e) θ.

altitude of approximately 20 km, has a spatial resolution of 18 m, including a total of 5211 pixels from 13 classes. The corresponding false-color image ground-truth map and ground-truth map are shown in Fig. 2(a) and (b), respectively. After removing water absorption and low SNR bands, 176 bands were used for the analysis.

The second dataset is the Houston dataset [60], which was acquired by the NSF-funded Center for Airborne Laser Mapping (NCALM) over the University of Houston campus and the neighboring urban area. The HSIry consists of 144 spectral bands with 349 × 1905 pixels in the 380–1050-nm region and has been calibrated to at-sensor spectral radiance units. A total of 15 029 pixels from 15 classes were used in this experiment. The corresponding false-color image ground-truth map and ground-truth map are shown in Fig. 3(a) and (b), respectively.

The third dataset is the Pavia University dataset [49], which is the part of the hyperspectral data of the image of the Italian city of Pavia by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in 2003. The ROSIS sensor is 0.43–0.86 continuous imaging of 115 bands in the μm wavelength range, and the resulting image has a spatial resolution of 1.3 m. Among them, 12 bands are eliminated due to the influence of noise, so the image formed by the remaining 103 spectral bands is generally used. The size of the data is 610 × 340 pixels, so it contains a total of 2 207 400 pixels, existing in a large number of background pixels with only 42 776 pixels containing features. These pixels contain a total of nine types of features. The corresponding false-color image ground-truth map and ground-truth map are shown in Fig. 4(a) and (b), respectively.

B. Extperimental Setting

In order to evaluate the effect of the proposed algorithm, we use the *1*-nearest neighbor (1NN) classifier to classify the data after DR and verify the effectiveness of the algorithm by the accuracy of the classification [34], [61]. We compare the proposed algorithm with four manifold learning algorithms, i.e., LFDA, LPP, NPE, and LMSDL, and three metric learning algorithms, i.e., LMNN, maximally collapsing metric learning (MCML) [62], and ITML, and the basic algorithm CMML. In order to ensure the fairness of the comparison, all methods used the same ground-truth data during the experiment. For each dataset, we randomly select 10% from each class as the training sample and the remaining data as the test sample. In order to reduce the error caused by random sampling, each experiment is repeated ten times, and the final result is the average of the ten experiments.

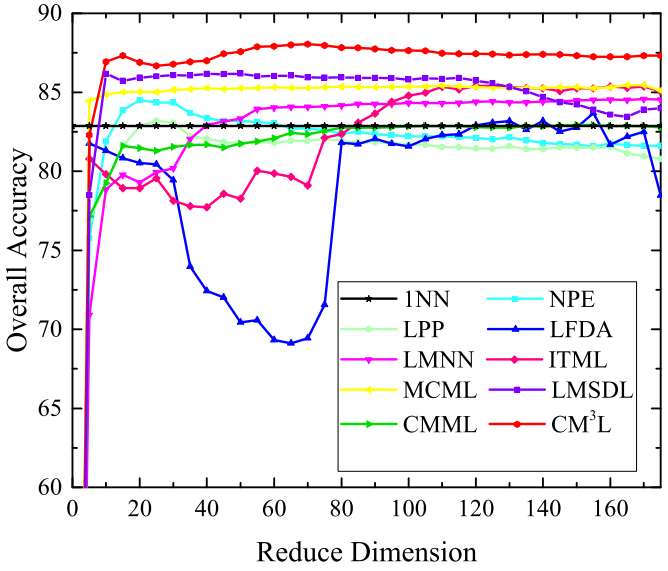


Fig. 6. Classification OA with respect to reduced dimensionality with the KSC dataset.

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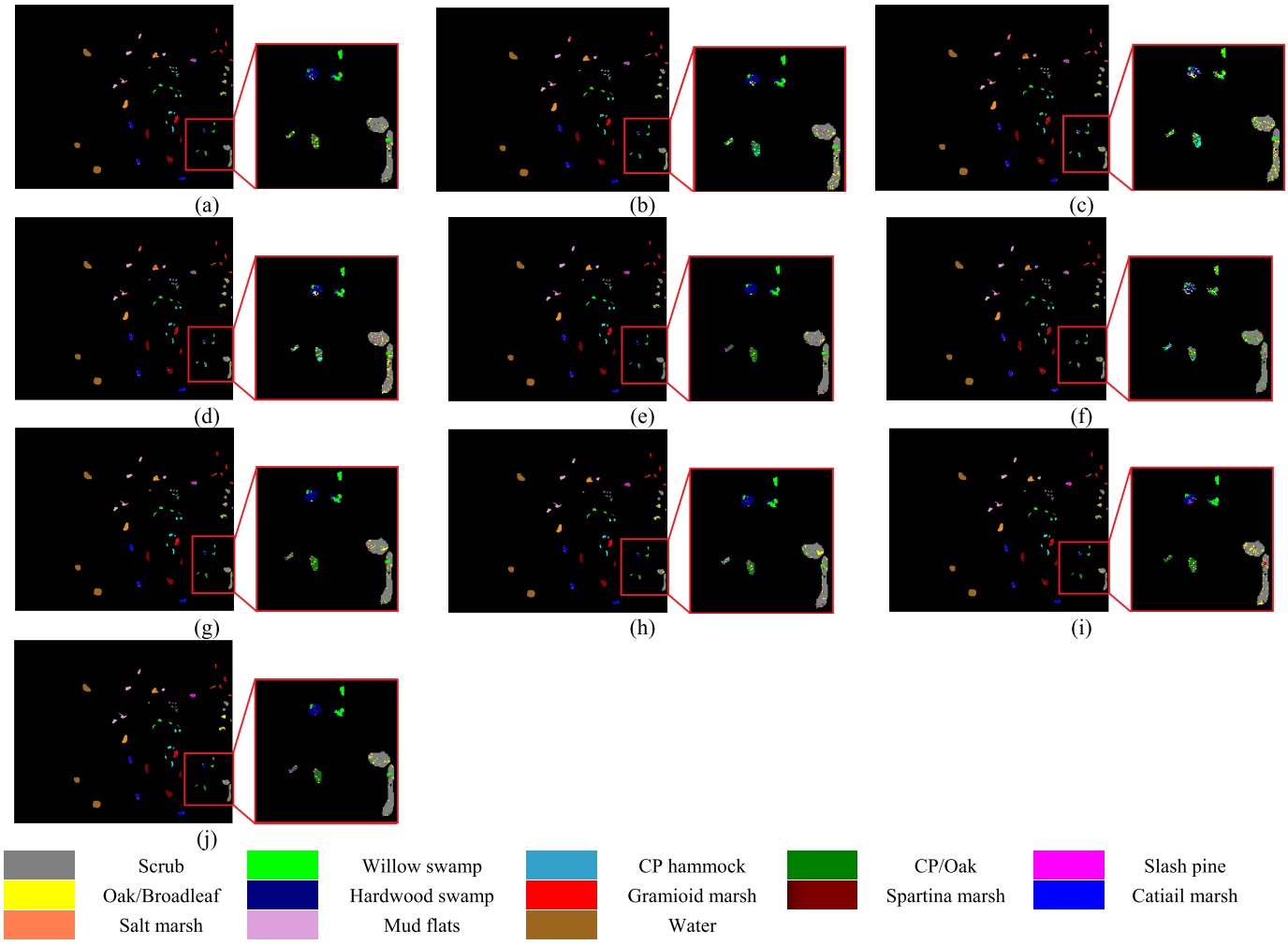


Fig. 7. Classification results for all the methods with the KSC dataset based on 1NN. (a) 1NN. (b) NPE. (c) LPP. (d) LFDA. (e) LMNN. (f) ITML. (g) MCML. (h) LMSDL. (i) CMML. (j) CM^{3}L.

For the proposed CM^{3}L algorithm, there are five parameters, including the number of neighboring points k, the number of clusters C, and three hyperparameters α, β, θ. As shown in Fig. 5, we can see that, in the value range of parameters k, C, α, θ, the final result is relatively stable and robust, so we set $k = 3, C = 3, \alpha = 0.0001$, and $\theta = 10$. As for parameter β, when its value is smaller, the result is better, so we set $\beta = 0.00001$.

C. Performance on the Three Datasets

1) KSC Dataset: The KCS dataset is divided into approximately 528 training pixels and 4683 test pixels. The numbers of samples for each class are listed in Table I. Fig. 6 presents the classification OA value of the 1NN classifier. From Fig. 6, we can see that, when the CM^{3}L algorithm becomes stable in ten dimensions, its accuracy is better than other algorithms, which proves the effectiveness of the proposed algorithm. The results of 1NN on the original KSC dataset are used as the baseline.

In Table II, the quantitative evaluation results of all the methods show more details, including the quantitative results of each class, the average OA and its standard deviation, the average kappa and its standard deviation, and the running time.

TABLE I
NUMBER OF SAMPLES OF THE KSC DATASET USED FOR EACH CLASS

No.	Class	Training	Test	No.	Class	Training	Test
C1	Scrub	77	684	C8	Grami-noid marsh	44	387
C2	Willow swamp	25	218	C9	Spar-tina marsh	52	468
C3	CP hamm-ock	26	230	C10	Catiail marsh	41	363
C4	CP/Oak	26	226	C11	Salt marsh	42	377
C5	Slash pine	17	144	C12	Mud flats	51	452
C6	Oak/Broad leaf	23	206	C13	Water	93	834
C7	Hard-wood swamp	11	94				
					Total	**528**	**4683**

The best results are bolded. From the results, we can find that CM^{3}L has the best classification accuracy in most classes of results, and CM^{3}L has the best OA and kappa. From the perspective of running time, CM^{3}L takes a little longer than the comparison algorithm, but it is also within an acceptable range.

Fig. 7 shows the classification maps of all considered methods for the KSC dataset, which gives a more direct presentation for classification results. From the figure, we can

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TABLE II
CLASSIFICATION ACCURACIES FOR THE KSC DATASET (DIM = 60)

Class	1NN	NPE	LPP	LFDA	LMNN	ITML	MCML	LMSDL	CMML	CM^{3}L
C1	89.33	76.61	80.12	71.05	86.11	88.60	84.94	86.26	90.94	**91.67**
C2	74.77	80.28	82.57	75.23	85.32	84.40	85.32	**85.78**	73.39	77.06
C3	75.65	58.70	68.26	58.26	77.39	72.61	80.87	79.57	76.09	**88.26**
C4	40.27	41.15	38.94	38.05	53.10	49.56	55.31	50.44	34.69	**62.39**
C5	45.14	41.67	49.31	42.36	59.72	66.67	61.11	**68.06**	46.53	58.33
C6	47.57	38.35	35.92	34.95	31.07	49.03	37.38	46.60	40.78	**59.22**
C7	82.98	79.79	79.79	72.34	89.36	84.04	88.30	86.17	80.85	**92.55**
C8	79.09	67.70	70.03	58.66	79.84	75.71	80.62	83.98	78.04	**81.65**
C9	96.58	92.95	94.23	90.38	94.23	90.17	95.30	96.37	92.31	**98.50**
C10	78.51	86.78	87.33	86.23	82.92	**90.63**	86.78	86.78	86.23	82.09
C11	90.19	89.66	90.19	86.21	**97.61**	93.90	97.08	96.29	87.00	90.72
C12	85.18	80.09	78.98	78.10	87.39	89.38	89.38	**90.71**	88.50	88.05
C13	99.88	97.96	97.96	97.84	98.08	98.08	97.96	97.84	99.88	**99.88**
OA	82.87 ±0.97	78.36 ±0.15	79.77 ±0.16	75.33 ±0.65	84.11 ±0.21	84.66 +±0.20	85.11 ±0.16	86.03 ±0.36	82.66 ±0.44	**87.12 ±0.43**
kappa	0.8091 ±0.0109	0.7596 ±0.0024	0.7752 ±0.0026	0.7262 ±0.0088	0.8233 ±0.0044	0.8294 ±0.0042	0.8345 ±0.0040	0.8445 ±0.0063	0.8066 ±0.0079	**0.8563 ±0.0076**
Time(s)	-	2.64	2.04	**1.27**	33	22	1010	19.9	25.78	41.15

TABLE III
NUMBER OF SAMPLES OF THE HOUSTON DATASET USED FOR EACH CLASS

No.	Class	Training	Test	No.	Class	Training	Test
C1	Healthy grass	126	1125	C9	Road	126	1129
C2	Stressed grass	126	1128	C10	Highway	123	1104
C3	Synthetic grass	70	627	C11	Railway	124	1111
C4	Trees	125	1119	C12	Parking Lot 1	124	1109
C5	Soil	125	1117	C13	Parking Lot 2	47	422
C6	Water	33	292	C14	Tennis Court	43	385
C7	Residential	127	1141	C15	Running Track	66	594
C8	Commercial	125	1119				
					Total	**1510**	**13519**

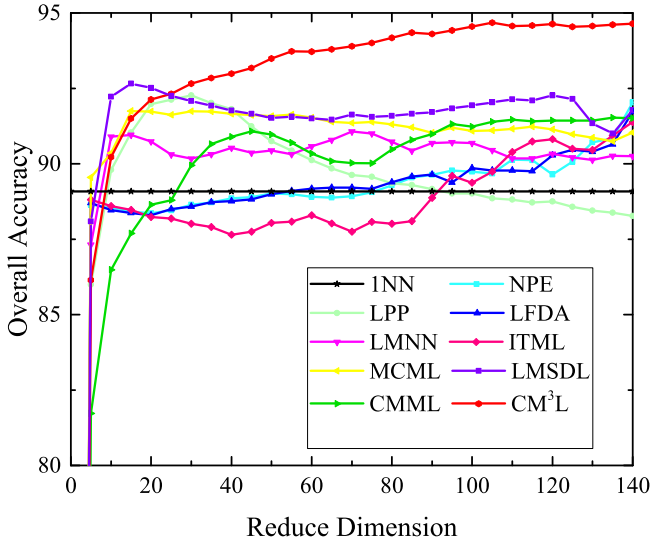


Fig. 8. Classification OA with respect to reduced dimensionality with the Houston dataset.

see that the proposed CM^{3}L algorithm has better classification results in most classes, especially for the regions of class 1 (Scrub), class 3 (CP hammock), and class 4 (CP/Oak). The results shown in Fig. 7 are consistent with the results shown in Table II.

2) Houston Dataset: The Houston dataset contains a total of 15 029 pixels, of which 1510 pixels are used as training samples, and the remaining 13 519 pixels are used as test samples. The numbers of samples for each class are listed in Table III.

In order to effectively illustrate the effectiveness of the proposed algorithm, Fig. 8 shows the classification OA values of all methods in different dimensions based on the 1NN classifier. It can be seen from Fig. 8 that, when the dimension is 20, the classification OA values of CM^{3}L are higher than other comparison algorithms, and as the dimension increases, the classification OA values of CM^{3}L are in a state of continuous increase, presenting the best classification OA.

The quantitative evaluation results of all the methods can explain the advantages of the proposed algorithm in more detail in Table IV. In general, CM^{3}L has the best OA and kappa, and it has more obvious advantages in most single classes; especially, the classification accuracy obtained by CM^{3}L in class 11 (Railway) and class 12 (Parking Lot 1) is much higher than the comparison algorithm. In addition, the average running time used by CM^{3}L to obtain the distance metric is also comparable with other algorithms.

As shown in Fig. 9, the proposed method achieves smoother classification maps than the other compared DR algorithms on the Houston dataset. It is obvious that CM^{3}L can achieve the best classification results in most of the ground cover classes, with the least error rate especially for the regions of class 7 (Residential), class 9 (Road), class 10 (Highway), class 11 (Railway), and class 12 (Parking Lot 1). The results shown in Fig. 9 are consistent with the results shown in Table IV.

3) Pavia University Dataset: The Pavia University dataset contains a total of 42 740 pixels in nine classes, of which the training sample has 4281 pixels, and the remaining 38 459 pixels are the test samples. More information is shown in Table V.

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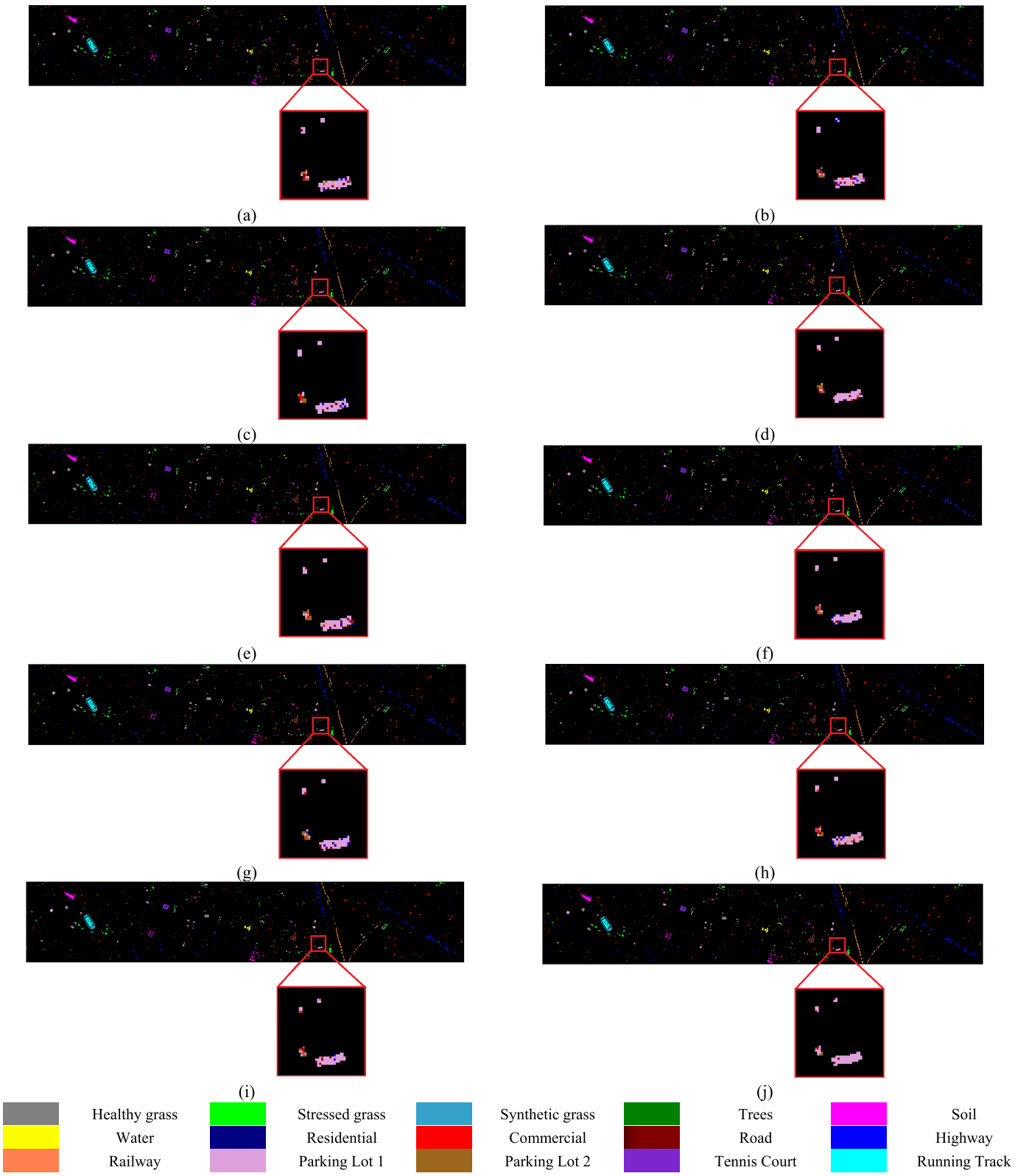


Fig. 9. Classification results for all the methods with the Houston dataset based on 1NN. (a) 1NN. (b) NPE. (c) LPP. (d) LFDA. (e) LMNN. (f) ITML. (g) MCML. (h) LMSDL. (i) CMML. (j) CM^{3}L.

Fig. 10 shows the classification OA values with regard to the reduced dimensionality for all the methods based on the 1NN classifier. From the figure, we can see that the OA values of CM^{3}L in the initial stage will be slightly lower than the NPE and LPP algorithms, but, when CM^{3}L stabilizes, CM^{3}L has the highest OA values. NPE and LPP have produced the Hughes phenomenon and the trend of OA decline.

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TABLE IV
CLASSIFICATION ACCURACIES FOR THE HOUSTON DATASET (DIM = 80)

Class	1NN	NPE	LPP	LFDA	LMNN	ITML	MCML	LMSDL	CMML	CM^{3}L
C1	98.40	98.40	97.87	98.40	**98.58**	97.33	97.69	96.98	98.31	97.16
C2	97.87	97.96	97.16	97.87	97.78	95.30	97.07	94.15	**98.05**	97.25
C3	98.56	98.56	99.68	98.56	99.52	98.81	99.52	**100**	99.68	99.84
C4	97.68	97.68	97.08	97.68	97.86	94.28	95.53	96.34	97.50	**97.68**
C5	96.06	96.42	97.76	96.24	96.06	96.60	97.05	**99.28**	96.42	98.84
C6	94.52	94.52	**98.29**	94.52	96.23	94.52	96.58	97.60	95.21	96.23
C7	86.59	87.03	87.64	86.94	90.10	84.14	89.13	87.99	87.82	**94.39**
C8	90.35	90.80	91.60	90.88	91.42	90.71	93.12	93.74	92.14	**94.91**
C9	77.71	78.33	79.66	78.24	80.02	78.60	82.68	86.94	79.31	**89.79**
C10	92.30	91.67	87.68	91.39	90.94	90.22	93.03	91.03	91.12	**96.65**
C11	84.43	84.43	83.98	84.52	87.22	84.97	86.86	83.44	87.31	**94.60**
C12	79.08	79.08	80.43	79.62	82.51	73.94	88.10	86.83	83.77	**92.25**
C13	27.49	29.62	24.88	32.46	**43.60**	33.89	38.39	41.47	37.20	41.94
C14	92.21	93.77	96.36	94.55	94.55	89.87	97.92	**98.96**	96.88	97.92
C15	99.83	99.83	99.83	99.83	99.83	99.66	99.83	99.83	98.83	**99.83**
OA	89.08 ±0.61	89.30 ±0.34	89.28 ±0.34	89.41 ±0.73	90.73 ±0.47	88.00 ±0.50	91.30 ±0.42	91.19 ±0.47	90.47 ±0.66	**94.17 ±0.56**
kappa	0.8818 ±0.0075	0.8842 ±0.0044	0.8841 ±0.0045	0.8854 ±0.0090	0.8997 ±0.0057	0.8708 ±0.0061	0.9059 ±0.0054	0.9047 ±0.0058	0.8969 ±0.0082	**0.9370 ±0.0069**
Time(s)	-	12.6	10.3	**1**	203	15.02	12405	7.84	80	90

TABLE V
NUMBER OF SAMPLES OF THE PAVIA UNIVERSITY DATASET USED FOR EACH CLASS

No.	Class	Training	Test	No.	Class	Training	Test
C1	Asphalt	664	5967	C6	Bare Soil	503	4526
C2	Meadows	1865	16784	C7	Bitumen	133	1197
C3	Gravel	210	1889	C8	Bricks	369	3313
C4	Trees	307	2757	C9	Shadows	95	852
C5	Metal sheets	135	1210				
					Total	**1510**	**13519**

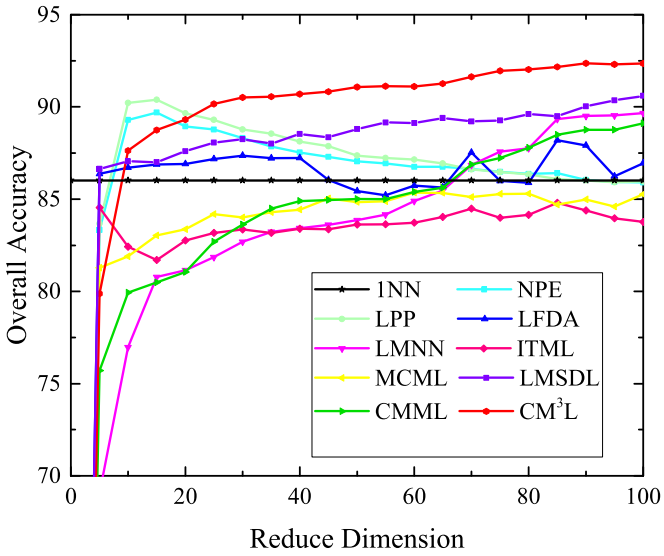


Fig. 10. Classification OA with respect to reduced dimensionality with the Pavia University dataset.

In Table VI, we use the quantitative evaluation results to make a more detailed and comprehensive comparison between proposed algorithms and comparison algorithms. From Table IV, we can see that CM^{3}L has the highest OA and kappa value, and CM^{3}L also has the best classification accuracy in most classes, including class 1 (Asphalt), class 3 (Gravel), class 4 (Trees), and class 6 (Bare Soil). Although the classification accuracy is not the best in other classes, there is only a slight gap from the highest accuracy. In terms of the running time, the running time of CM^{3}L is higher than the three algorithms of manifold learning. This is because CM^{3}L needs to calculate every set of constraint pairs in the training sample. However, combined with the classification accuracy results, CM^{3}L still has certain advantages.

In order to further demonstrate the effectiveness of the CM^{3}L algorithm, Fig. 11 shows the classification maps based on different algorithms, which are consistent with the results shown in Table VI. From the classification maps, we can see that the classification result of CM^{3}L has obvious advantages on the Pavia University dataset, especially in the case of Gravel and Bricks, which have fewer training samples. At the same time, we can find that most of the comparison algorithms are easy to misclassify Meadows and Bare Soil, and the proposed CM^{3}L can distinguish these two types of ground objects.

V. CONCLUSION

This article proposes a metric learning algorithm CM^{3}L for DR of HSI. CM^{3}L introduces the idea of manifold learning on the basis of traditional metric learning. CM^{3}L constructs the sample point and its surrounding neighboring points into a manifold and then calculates the distance between the manifold and the manifold to replace the traditional Mahalanobis distance, so as to enhance the discriminant ability by exploiting intrinsic structure information. Through experiments on extensive HSI datasets, the proposed CM^{3}L algorithm can obtain better experimental results compared with traditional DR and the most advanced metric learning methods. Our future work will try to incorporate spatial information into the distance calculation process and introduce the proposed manifold distance into other metric learning, testing the effectiveness of this distance through experiments.

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TABLE VI

CLASSIFICATION ACCURACIES FOR THE PAVIA UNIVERSITY DATASET (DIM = 60)

Class	1NN	NPE	LPP	LFDA	LMNN	ITML	MCML	LMSDL	CMML	CM^3L
C1	86.68	84.16	86.21	78.85	86.27	84.63	86.56	88.20	86.88	**89.64**
C2	93.53	95.32	95.25	95.83	91.93	90.40	92.96	**96.83**	92.15	96.63
C3	67.39	62.26	64.16	71.52	71.57	52.25	66.17	70.09	74.43	**78.88**
C4	83.53	85.38	84.55	86.25	85.67	84.91	83.06	88.21	86.22	**90.39**
C5	99.09	**99.75**	99.59	99.67	99.50	97.85	99.34	99.34	99.50	99.01
C6	63.57	68.98	69.18	73.71	60.87	60.87	61.64	69.82	61.11	**82.79**
C7	80.03	83.71	81.87	**93.40**	77.19	71.43	83.38	81.87	78.61	83.21
C8	82.31	82.10	78.36	**83.10**	80.26	70.39	81.71	85.99	80.89	82.46
C9	**100**	100	100	99.88	99.77	99.88	100	99.77	99.88	99.88
OA	85.87 ±0.34	86.90 ±0.29	86.86 ±0.27	87.75 ±0.66	84.90 ±0.55	81.89 ±0.59	85.35 ±0.37	89.13 ±0.41	85.40 ±0.36	**91.11 ±0.34**
kappa	0.8109 ±0.0050	0.8247 ±0.0035	0.8240 ±0.0033	0.8368 ±0.0117	0.7984 ±0.0095	0.7595 ±0.0104	0.8039 ±0.0055	0.8542 ±0.0064	0.8051 ±0.0054	**0.8817 ±0.0051**
Time(s)	-	284	145	**1**	23191	11	15230	63	1377	926

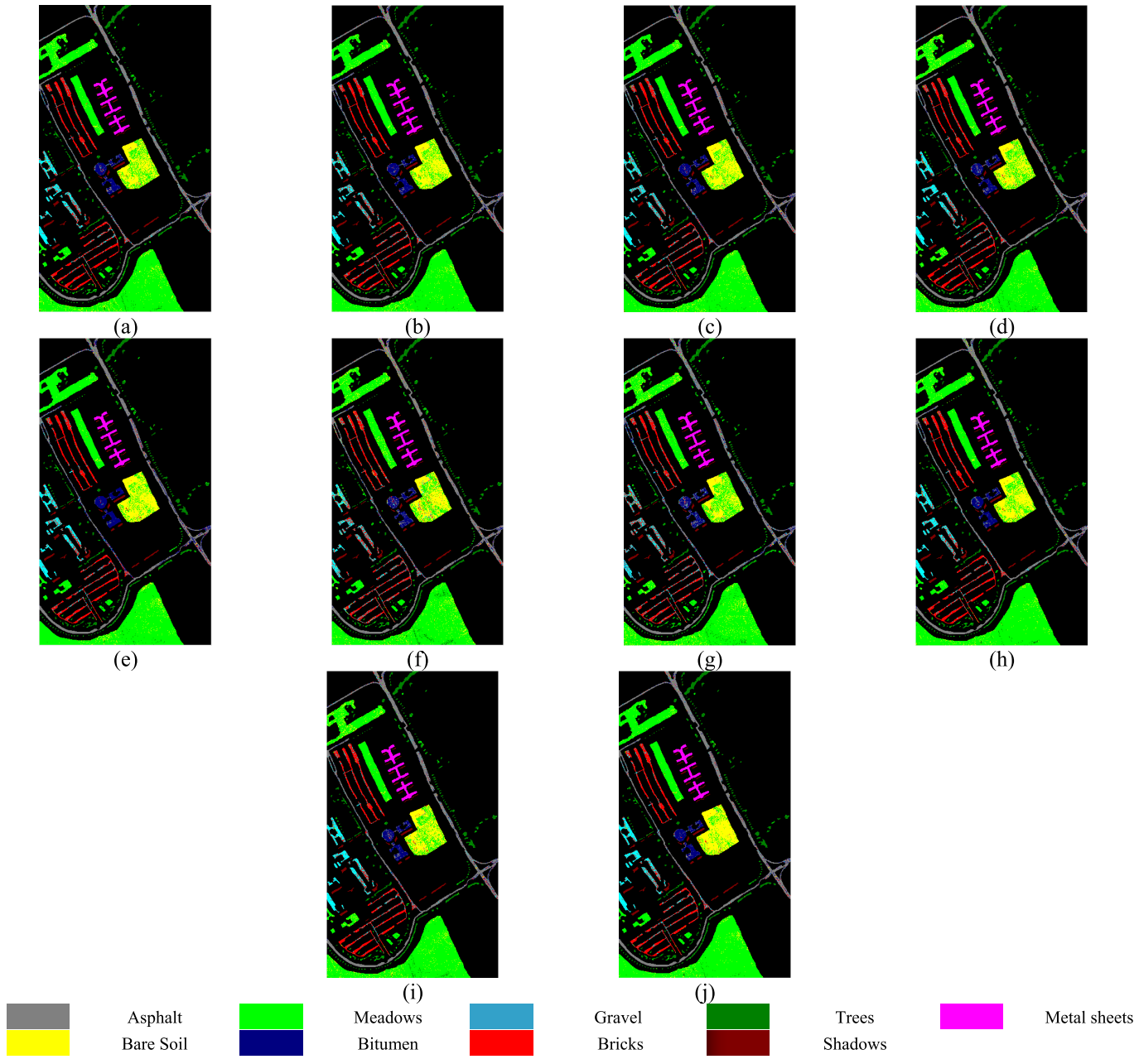


Fig. 11. Classification results for all the methods with the Pavia University dataset based on 1NN. (a) 1NN. (b) NPE. (c) LPP. (d) LFDA. (e) LMNN. (f) ITML. (g) MCML. (h) LMSDL. (i) CMML. (j) CM3L.

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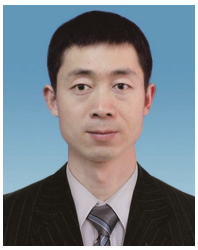
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